Cover Letter with Heavy Math, as Requested by Teagan Tomesh

October 29, 2026

How to Read This Cover Letter:

1. All the big-picture ideas are in the first section.
2. If the math is boring, just read the bold text.

1 The normal cover letter.

Dear Hiring Manager,

My name is Maureen Zhang. I completed my PhD in Mathematics at the University of California, Irvine in March 2026. By training, I am a mathematician specializing in noncommutative ring theory and category theory. During **IEEE Quantum Week** in Toronto about a month ago, I had the pleasure of speaking with engineer **Dr.Teague Tomesh.** We had a nice conversation, and he encouraged me to apply for this position. He suggested that I explain why my research would be helpful for this role and mention his name.

I am applying for the **Quantum Software Engineer** position because I am passionate about the future of quantum computing and I love working with people. I believe I can contribute meaningfully to Infleqion.

Sincerely,
Maureen Zhang

So next we have the not-so-conventional cover letter!

2 Details of My Math, If You Care to Read

Disclaimer: I wrote this draft myself. Just because it's long doesn't mean it's AI slop! AI was used to check the grammar and help insert and format the mathematical equations. I reviewed every line, and the mathematical content and final wording are my responsibility.

3 I Have a Chain-Complex-Native Brain

I spent years studying homological algebra and category theory. So when I came across Kitaev's toric code, the X- and Z-checks immediately looked like boundary and coboundary maps to me.

$$0 \longrightarrow \text{plaquettes } (Z\text{-checks}) \xrightarrow{\partial_2} \text{edges (physical qubits)} \xrightarrow{\partial_1} \text{vertices } (X\text{-checks}) \longrightarrow 0.$$

Moreover, category theory training has helped me greatly in understanding lift products as well.

3.1 For the Math friend to read

One useful approach to constructing qLDPC codes and CSS code is through chain complexes.

Let

$$C_\bullet : \quad C_2 \xrightarrow{\partial_2} C_1 \xrightarrow{\partial_1} C_0$$

be a three-term chain complex of finite-dimensional vector spaces over $\mathbb{F}_2$, with

$$\partial_1 \circ \partial_2 = 0.$$

After choosing bases, we obtain a CSS quantum code by defining its parity-check matrices as

$$\boxed{H_X = \partial_1, \qquad H_Z = \partial_2^{\mathsf{T}}.}$$

The chain-complex condition becomes

$$\boxed{H_X H_Z^{\mathsf{T}} = \partial_1 \partial_2 = 0,}$$

which is precisely the commutativity condition for the X- and Z-type stabilizers.

In this language, the physical qubits correspond to basis elements of C_1, while the X- and Z-type stabilizer generators correspond to basis elements of C_0 and C_2, respectively.

The number of physical qubits is

$$n = \dim_{\mathbb{F}_2} C_1.$$

More importantly, the number of encoded logical qubits is determined by the first homology group:

$$\boxed{H_1(C_\bullet) = \frac{\ker \partial_1}{\operatorname{im} \partial_2}, \qquad k_{\text{logical}} = \dim H_1(C_\bullet).}$$

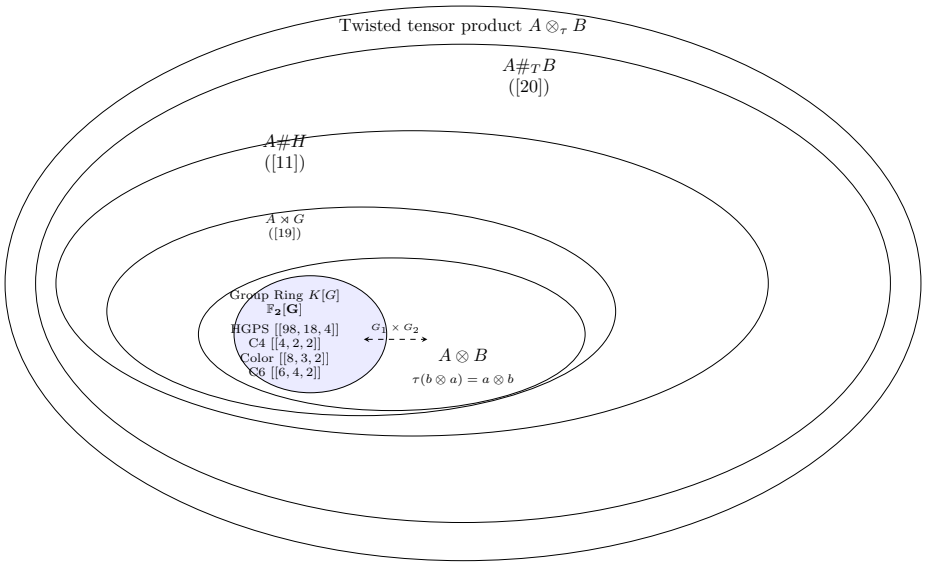
Thus, logical operators are naturally described by cycles that are not boundaries, or cocycles that are not coboundaries.This motivates a homological formulation of the code-construction problem.

For me personally, this formalization made understanding and picking up QEC feel like running into your own mother in a foreign country.

4 My research training has made me looking at all the current QEC from a even more generalized angle

In quantum error correction (QEC), taking tensor products is a way to generate new codes.

My Ph.D. research focused on a **generalization** of the tensor product called the **twisted tensor product**, which encodes more complex "gluing conditions." There relationship is viewed as follow: (detailed definition of each ring see table 1 at section 7.



And just like we can take the twisted tensor product of two algebras A and B and get $A \otimes_\tau B$, we can study twisted tensor products of differential graded algebras(i.e. chains) too, and get something like

$$\boxed{C_\bullet \otimes_{\widetilde{\tau}} D_\bullet,}$$

where $\widetilde{\tau}$ is a twisting map compatible with the differentials.

My reseach devoted my Ph.D. to studying in what situations the Ext-algebra of a twisted tensor product is again a twisted tensor product. I studied the homological algebra of twisted tensor products. **So heuristically, think of $E(A)$ as homological information attached to A, but with an algebra structure. And I studied under what specific conditions we would have**

$$\boxed{E(A \otimes_\tau B) \cong E(B) \otimes_{\tau^!} E(A).}$$

In particular, my theorem considers Koszul algebras A and B with a strongly graded, invertible twisting map τ, under the hypotheses of my dissertation. Here $\tau^!$ is the induced twisting map on their Ext-algebras. **All the popular codes on the market can be easily understood through this framework.**

5 More Math: Skip if You're Not Interested, Thank you for reading :)

The following section is partly to show off that I do know QEC codes, and partly to give the reader an idea of how I understand them! You can skip them all, all the condensed information are in section table and charts. 7. I grouped the codes by its underlying algebra, one sections for bland tensor products 5.1, and one section for twisted tensor products with non-trivial twist 5.2.

5.1 Examples of Codes as tensor product

Iceberg code $[[4, 2, 2]]$**.** Let

$$A = \frac{\mathbb{F}_2[x]}{(x^2 - 1)}, \qquad B = \frac{\mathbb{F}_2[y]}{(y^2 - 1)}.$$

Then

$$R = A \otimes_{\tau_{\text{flip}}} B,$$

where

$$\boxed{\tau_{\text{flip}}(y^j \otimes x^i) = x^i \otimes y^j.}$$

So this is just the ordinary tensor product.
Let

$$u = 1 + x, \qquad v = 1 + y, \qquad I = (u, v) \subseteq R.$$

Now we have a short exact sequence of $\mathbb{F}_2$-vector spaces $C_\bullet$

$$\boxed{0 \longrightarrow \mathbb{F}_2 \xrightarrow{\iota_A} A \xrightarrow{\varepsilon_A} \mathbb{F}_2 \longrightarrow 0,}$$

where the maps are explicitly given by

$$\boxed{\begin{aligned} \iota_A(1) &= 1 + x = u, \\ \varepsilon_A(a_0 + a_1 x) &= a_0 + a_1. \end{aligned}}$$

And similarly, for B, $D_\bullet$

$$\boxed{0 \longrightarrow \mathbb{F}_2 \xrightarrow{\iota_B} B \xrightarrow{\varepsilon_B} \mathbb{F}_2 \longrightarrow 0,}$$

with

$$\boxed{\begin{aligned} \iota_B(1) &= 1 + y = v, \\ \varepsilon_B(b_0 + b_1 y) &= b_0 + b_1. \end{aligned}}$$

Notice that

$$\varepsilon_A \circ \iota_A = 0, \qquad \varepsilon_B \circ \iota_B = 0.$$

In fact, both sequences are exact.

Now, by the Künneth formula, the ordinary tensor product of these two chain complexes has

$$\boxed{H_n(C_\bullet \otimes_{\mathbb{F}_2} D_\bullet) \cong \bigoplus_{p+q=n} H_p(C_\bullet) \otimes_{\mathbb{F}_2} H_q(D_\bullet) = 0.}$$

So the ordinary tensor-product complex is exact too.

But for the Iceberg code, we can take the degreewise tensor product of the two sequences instead.

That gives the chain complex

$$\boxed{0 \longrightarrow \mathbb{F}_2 \otimes \mathbb{F}_2 \xrightarrow{\partial_2} A \otimes B \xrightarrow{\partial_1} \mathbb{F}_2 \otimes \mathbb{F}_2 \longrightarrow 0,}$$

where

$$\boxed{\partial_2 = \iota_A \otimes \iota_B, \qquad \partial_1 = \varepsilon_A \otimes \varepsilon_B.}$$

The differentials are explicitly

$$\boxed{\begin{aligned} \partial_2(1) &= (1 + x) \otimes (1 + y) \\ &= 1 + x + y + xy, \end{aligned}}$$

and

$$\boxed{\partial_1(x^i y^j) = 1, \qquad i, j \in \{0, 1\}.}$$

Here we identify $\mathbb{F}_2 \otimes \mathbb{F}_2$ with $\mathbb{F}_2$. Thus,

$$\boxed{0 \longrightarrow \mathbb{F}_2 \xrightarrow{\cdot(1+x)(1+y)} R \xrightarrow{\varepsilon} \mathbb{F}_2 \longrightarrow 0.}$$

Here the first arrow sends 1 to $(1 + x)(1 + y)$.

And because

$$\varepsilon((1 + x)(1 + y)) = 0,$$

over characteristic 2, we have

$$\boxed{\partial_1 \partial_2 = 0.}$$

This concludes the parity check !

Moreover, because the kernel of $\partial_1 = \varepsilon$ is the ideal generated by $1 + x$ and $1 + y$,

$$\boxed{\ker \partial_1 = I = (1+x, 1+y),}$$

and

$$I^2 = \mathbb{F}_2 \cdot (1+x)(1+y),$$

we have

$$\boxed{\mathbb{F}_2 \cong I^2 \quad \text{as } \mathbb{F}_2\text{-modules.}}$$

$$\boxed{\begin{array}{l}\phi : \mathbb{F}_2 \xrightarrow{\sim} I^2, \\ \qquad 1 \longmapsto (1+x)(1+y).\end{array}}$$

We thus have an isomorphism of chain complexes

$$\boxed{\left(0 \longrightarrow \mathbb{F}_2 \xrightarrow{\partial_2} R \xrightarrow{\varepsilon} \mathbb{F}_2 \longrightarrow 0\right) \cong \left(0 \longrightarrow I^2 \xhookrightarrow{\iota} R \xrightarrow{\varepsilon} \mathbb{F}_2 \longrightarrow 0\right)}$$

in the category Ch(R-Mod). Thus the two chain complexes are isomorphic, not merely quasi-isomorphic. Consequently, their first homology groups are isomorphic.

This establishes the chain-complex formulation of the $[[4, 2, 2]]$ Iceberg code.

C6 code $[[6, 4, 2]]$**.** Take

$$A = \frac{\mathbb{F}_2[x]}{(x^2-1)}, \qquad B = \frac{\mathbb{F}_2[y]}{(y^3-1)}.$$

Then

$$R = A \otimes B$$

.

The algebra R is also isomorphic to $\mathbb{F}_2[z]/(z^6-1)$.

Let

$$N = 1 + z + z^2 + z^3 + z^4 + z^5.$$

The complex

$$0 \longrightarrow \langle N \rangle \hookrightarrow R \xrightarrow{\varepsilon} \mathbb{F}_2 \longrightarrow 0$$

gives the $[[6, 4, 2]]$ code.

Eight-qubit CSS color code $[[8,3,2]]$**.** Take

$$A = \frac{\mathbb{F}_2[x]}{(x^2-1)},$$

$$B = \frac{\mathbb{F}_2[y,z]}{(y^2-1, z^2-1)}.$$

Then

$$R = A \otimes_\tau B,$$

Let

$$I = (1+x, 1+y, 1+z) \subseteq R.$$

The augmentation-ideal complex

$$0 \longrightarrow I^2 \hookrightarrow R \xrightarrow{\varepsilon} \mathbb{F}_2 \longrightarrow 0$$

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Many-hypercube $[[16,4,4]]$**.** Now let's consider the $[[16,4,4]]$ many-hypercube code, which is related to the $[[4,2,2]]$ Iceberg code through concatenation.

Again, we start with two algebras whose generators have finite order:

$$A = \frac{\mathbb{F}_2[x,y]}{(x^2-1, y^2-1)},$$

$$B = \frac{\mathbb{F}_2[p,q]}{(p^2-1, q^2-1)}.$$

We take their ordinary tensor product

$$\boxed{R = A \otimes B,}$$

Specifically,

$$\boxed{R \cong \frac{\mathbb{F}_2[x,y,p,q]}{(x^2-1, y^2-1, p^2-1, q^2-1)}.}$$

Now let

$$u = 1+x, \qquad v = 1+y,$$

and define two elements of B:

$$P = 1+pq, \qquad Q = p+q.$$

Define the ideal

$$\boxed{J = (uv,\ uP+vQ,\ vP+uQ) \subseteq R.}$$

We can write J explicitly as an $\mathbb{F}_2$-vector space:

$$\boxed{J = \mathrm{span}_{\mathbb{F}_2}\big\{uv,\ puv,\ quv,\ pquv,\ uP + vQ,\ vP + uQ\big\}.}$$

Furthermore, using $u^2 = v^2 = P^2 = Q^2 = PQ = 0$, we obtain

$$\boxed{J^2 = 0.}$$

This is exactly the algebraic condition we need to construct our chain complex.

The chain complex.

Define the linear functional

$$\lambda : R \longrightarrow \mathbb{F}_2$$

by chopping off all nonconstant terms and keeping only the constant term in the polynomial .

For any $r, s \in R$, let

$$\langle r, s\rangle = \lambda(rs).$$

Now consider the chain complex

$$\boxed{0 \longrightarrow J \xhookrightarrow{\partial_2} R \xrightarrow{\partial_1} J^* \longrightarrow 0,}$$

where

$$J^* = \mathrm{Hom}_{\mathbb{F}_2}(J, \mathbb{F}_2).$$

The first differential is inclusion:

$$\boxed{\partial_2(j) = j.}$$

The second differential is explicitly

$$\boxed{\begin{aligned}&\partial_1 : R \longrightarrow J^*,\\ &\qquad r \longmapsto \big(j \longmapsto \lambda(rj)\big).\end{aligned}}$$

Because $J^2 = 0$, for any $j, j' \in J$,

$$\partial_1(\partial_2(j))(j') = \lambda(jj') = 0.$$

Thus,

$$\boxed{\partial_1\partial_2 = 0.}$$

We have constructed the many-hypercube code starting from the tensor product of two algebras and its generating relations to obtain

$$\boxed{0 \longrightarrow J \longrightarrow R \longrightarrow J^* \longrightarrow 0.}$$

Its logical qubits are described by

$$\boxed{H_1 \cong \frac{\mathrm{Ann}_R(J)}{J}.}$$

Knill's C6 code $[[6, 2, 2]]$**.** Let

$$A = \frac{\mathbb{F}_2[x]}{(x^2 - 1)}, \qquad B = \frac{\mathbb{F}_2[y]}{(y^3 - 1)}.$$

Then

$$R = A \otimes B.$$

Define the ideal

$$\boxed{J = ((1 + x)(1 + y)) \subseteq R.}$$

Now define

$$\boxed{0 \longrightarrow J \xhookrightarrow{\partial_2} R \xrightarrow{\partial_1} J^* \longrightarrow 0,}$$

where

$$\partial_1(r)(j) = \lambda(rj^*).$$

$$\partial_2(j) = j$$

. Then

$$\partial_1(r)(j) = \lambda(rj^*).$$

Since J is stable under $*$ and $J^2 = 0$, we obtain

$$\boxed{\partial_1\partial_2 = 0.}$$

giving the code

$$\boxed{[[6, 2, 2]].}$$

5.2 Examples of Codes with Nontrivial Twisted Tensor Product Structures

Non-Abelian two-block group-algebra (2BGA) codes. Another example is the non-Abelian 2BGA code. Let

$$A = \frac{\mathbb{F}_2[x]}{(x^3 - 1)}, \qquad B = \frac{\mathbb{F}_2[y]}{(y^2 - 1)}.$$

This time, we take a nontrivial twisted tensor product

$$\boxed{R = A \otimes_\tau B,}$$

where

$$\boxed{\tau(y^j \otimes x^i) = x^{(-1)^j i} \otimes y^j.}$$

So $yx = x^{-1}y$, and the algebra is noncommutative. For $a, b \in R$, define

$$\boxed{H_X = [L(a) \mid R(b)], \qquad H_Z = [R(b)^T \mid L(a)^T].}$$

Here $L(a)$ and $R(b)$ denote left and right multiplication, respectively. Because left and right multiplication commute, we have $H_X H_Z^T = 0$. We can actually write out the chain complex:

$$\boxed{0 \longrightarrow R \xrightarrow{\begin{bmatrix} R(b) \\ L(a) \end{bmatrix}} R^2 \xrightarrow{\begin{bmatrix} L(a) & R(b) \end{bmatrix}} R \longrightarrow 0.}$$

Moreover, this twist satisfies the compatibility conditions in Definition 1.

Eight-qubit Gottesman code $[[8, 3, 3]]$**.** Another interesting example is the eight-qubit Gottesman code, which is non-CSS.

Over $\mathbb{C}$, let

$$A = \mathbb{C}[C_2^8], \qquad B = \mathbb{C}[C_2^8],$$

where A represents the X operators and B represents the Z operators. The underlying Pauli operator algebra is indeed a twisted tensor product

$$\boxed{P = A \otimes_\tau B,}$$

where the twisting map is

$$\boxed{\tau(z^{\mathbf{v}} \otimes x^{\mathbf{u}}) = (-1)^{\mathbf{u}\cdot\mathbf{v}} x^{\mathbf{u}} \otimes z^{\mathbf{v}}.}$$

In particular,

$$\boxed{z_i x_i = -x_i z_i.}$$

This is a genuinely nontrivial twisting map, satisfying the unit and compatibility conditions in Definition 1.

With special choice of H and J, we can still associate a chain complex to the code:

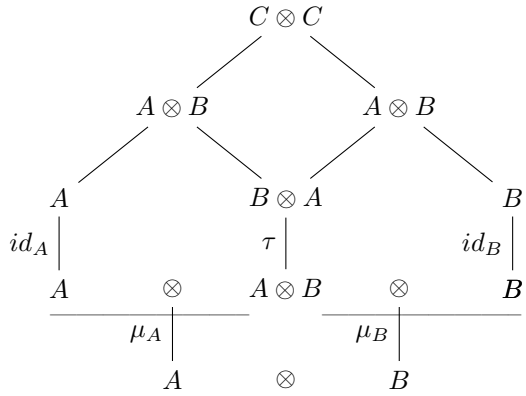
$$\boxed{0 \longrightarrow \mathbb{F}_2^5 \xrightarrow{H^T} \mathbb{F}_2^{16} \xrightarrow{HJ} \mathbb{F}_2^5 \longrightarrow 0.}$$

6 What is a twisted tensor product

During my Ph.D., I studied a construction in noncommutative algebra called the **twisted tensor product**, denoted $A \otimes_\tau B$, where τ is a map from $B \otimes A$ to $A \otimes B$:

$$\tau : B \otimes A \longrightarrow A \otimes B.$$

It generalizes the ordinary tensor product. If we denote $C = A \otimes_\tau B$, its multiplication looks like the diagram below, where the diagram is read from top to bottom.



For a regular tensor, the τ will be the swap map between the two vector space A and B.

The formal definition of such a twist is as follows.

Definition 1. Let (A, μ_A) and (B, μ_B) be k-algebras, where μ_A and μ_B denotes the multiplications on A and B respectivly. A k-linear map

$$\tau : B \otimes A \longrightarrow A \otimes B$$

is an **algebra twist** if and only if $\tau(b \otimes 1) = 1 \otimes b$, $\tau(1 \otimes a) = a \otimes 1$, and the following two compatibility condition is satisfied:

$$(1_A \circ \mu_B) \circ (\tau \otimes 1_A) \circ (1_B \otimes \tau) = \tau \circ (\mu_B \otimes 1_A) \tag{1}$$

$$(\mu_A \circ 1_B) \circ (1_A \otimes \tau) \circ (\tau \otimes 1_A) = \tau \circ (1_B \otimes \mu_A) \tag{2}$$

7 Tables and Graphs

Construction / code	**Multiplication**	**Twisting map** $\tau\colon B\otimes A\to A\otimes B$
I. Algebra constructions		
Twisted tensor product $A\otimes_\tau B$	If $\tau(b\otimes a')=\sum_i a_i'\otimes b_i$, then $(a\otimes b)(a'\otimes b')=\sum_i aa_i'\otimes b_ib'$.	$\tau(b\otimes a)=\sum_i a_i\otimes b_i$, subject to the twisting identities.
Ordinary tensor product $A\otimes B$	$(a\otimes b)(a'\otimes b')=aa'\otimes bb'$.	$\tau(b\otimes a)=a\otimes b$ (the flip).
Hopf smash product $A\#H$	For $\Delta(h)=\sum h_{(1)}\otimes h_{(2)}$, $(a\#h)(a'\#h')=\sum a\,(h_{(1)}\cdot a')\#h_{(2)}h'$.	$\tau(h\otimes a)=\sum(h_{(1)}\cdot a)\otimes h_{(2)}$.
Takeuchi smash product $A\#B$	With A an H-module algebra and B an H-comodule algebra, $\rho(b)=\sum b_{(-1)}\otimes b_{(0)}$, $(a\#b)(a'\#b')=\sum a\,(b_{(-1)}\cdot a')\#b_{(0)}b'$.	$\tau(b\otimes a)=\sum(b_{(-1)}\cdot a)\otimes b_{(0)}$.
Ore extension $A[x;\sigma,\delta]$	Multiplication is determined by $xa=\sigma(a)x+\delta(a)$ and multiplication in A.	Set $B=k[x]$: $\tau(x\otimes a)=\sigma(a)\otimes x+\delta(a)\otimes 1$.
Direct-product group algebra $k[N\times Q]$	$(n\otimes q)(n'\otimes q')=nn'\otimes qq'$	$A=k[N],\ B=k[Q]$ $\tau(q\otimes n)=n\otimes q$
Semidirect-product group algebra $k[N\rtimes_\varphi Q]$	$(n\otimes q)(n'\otimes q')=n\varphi_q(n')\otimes qq'$	$A=k[N],\ B=k[Q]$ $\boxed{\tau(q\otimes n)=\varphi_q(n)\otimes q}$

Throughout, let $k=\mathbb{F}_2$ for all the QEC codes.

Table 3: Group-Algebra Realizations of CSS Codes: Trivial Twisting Maps

Name of Code	**Underlying Algebra**	**As a Tensor Product**
HGPS $[[98,18,4]]$	$R=k[C_7\times C_7]$.	$R\cong A\otimes_k B$, where $A=k[x]/(x^7-1)$, $B=k[y]/(y^7-1)$.

Name of Code	**Underlying Algebra**	**As a Tensor Product**
Bivariate bicycle (BB) codes	$R = k[C_\ell \times C_m]$.	$$R \cong A \otimes_k B,$$ where $$A = k[x]/(x^\ell - 1),$$ $$B = k[y]/(y^m - 1).$$
C4 / Iceberg $[[4, 2, 2]]$	$R = k[C_2^2]$. Let $$I = (1 + x, 1 + y).$$ The chain complex is $$0 \longrightarrow I^2 \hookrightarrow R \xrightarrow{\varepsilon} k \longrightarrow 0.$$	$$R \cong A \otimes_k B,$$ where $$A = k[x]/(x^2 - 1),$$ $$B = k[y]/(y^2 - 1).$$
Six-qubit Iceberg $[[6, 4, 2]]$	$R = k[C_6]$.	Since $C_6 \cong C_2 \times C_3$, $$R \cong A \otimes_k B,$$ where $$A = k[x]/(x^2 - 1),$$ $$B = k[y]/(y^3 - 1).$$
Eight-qubit color code $[[8, 3, 2]]$	$R = k[C_2^3]$. Let $$I = (1 + x, 1 + y, 1 + z).$$ The chain complex is $$0 \longrightarrow I^2 \hookrightarrow R \xrightarrow{\varepsilon} k \longrightarrow 0.$$	$$R \cong A \otimes_k B \otimes_k C,$$ where $$A = k[x]/(x^2 - 1),$$ $$B = k[y]/(y^2 - 1),$$ $$C = k[z]/(z^2 - 1).$$
Many-hypercube $[[16, 4, 4]]$	$R = k[C_2^4]$.	$$R \cong A \otimes_k B,$$ where $$A = \frac{k[x, y]}{(x^2 - 1, y^2 - 1)},$$ $$B = \frac{k[s, t]}{(s^2 - 1, t^2 - 1)}.$$

Table 4: Code with a Nontrivial Twisting Map

Name of Code	**Underlying Algebra**	**As a Twisted Tensor Product**
Eight-qubit Gottesman $[[8,3,3]]$	Non-CSS: stabilizers mix X, Y, and Z. For $a, b, a', b' \in \mathbb{F}_2^8$, $$(X^a Z^b)(X^{a'} Z^{b'}) = (-1)^{b \cdot a'} X^{a+a'} Z^{b+b'}.$$	Over $\mathbb{C}$, let $$A = \mathbb{C}[C_2^8] \quad (X\text{-operators}),$$ $$B = \mathbb{C}[C_2^8] \quad (Z\text{-operators}).$$ Then $$\boxed{R = A \otimes_\tau B,}$$ where $$\boxed{\begin{aligned}&\tau : B \otimes A \longrightarrow A \otimes B,\\ &\tau(Z^b \otimes X^a)\\ &\quad = (-1)^{a \cdot b} X^a \otimes Z^b.\end{aligned}}$$
Non-Abelian 2BGA $[[12,2,2]]$	Let $$A = \frac{\mathbb{F}_2[x]}{(x^3-1)}, \qquad B = \frac{\mathbb{F}_2[y]}{(y^2-1)}.$$	The nontrivial twisting map is $$\boxed{\tau(y^j \otimes x^i) = x^{(-1)^j i} \otimes y^j.}$$ i.e. $$yx = x^{-1}y.$$

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